DP Graded Quiz M3

Question 1

If $dX_t = 2dt + 5dW_t$ and $Y_t = X_t^2$, then $d[Y, Y]_t$ is equal to:

- $10X_t dt$
- $100X_t^2 dt$
- $100dt$
- $X_t^2 dt$

Question 2

A market with a unique equivalent Martingale measure:

- is arbitrage-free and complete.
- is not arbitrage-free and is incomplete.
- is arbitrage-free and incomplete.
- is not arbitrage-free and is complete.

Question 3

If X_t follows an Ornstein-Uhlenbeck process with parameters $a = 1$ and $\sigma = 0.1$, compute the variance of X_∞.

- 0.01
- 0.1
- 0.005
- 0.25

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Question 4

If an Ornstein-Uhlenbeck process with parameters a, $sigma$ and c, the parameter c represents _____.

- the drift
- a constant representing the mean that the process reverts to
- the speed of mean reversion
- the volatility

Question 5

If $dX_t = dt + 2t^4 dW_t$, compute the quadratic variation of $[X, X]_t$.

- $5t^9$
- $\frac{5}{9}t^7$
- $11t^8$
- $\frac{4}{9}t^9$

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Question 6

In a trinomial tree with $\sigma = 0.5$, $T = 1$, and $S_0 = 50$, compute the price of a European call option with $K = 90$, $N = 100$, and $r = 0$.

- 1.938
- 23.221
- 19.236
- 0

Question 7

If X_t follows an Ornstein-Uhlenbeck process with parameters $a = 10$ and $\sigma = 0.5$, compute the variance of X_∞.

- 0.0125
- 0.025
- 1.25
- 0.125

Question 8

If X_t follows an Ornstein-Uhlenbeck process with $a = 10$, compute the expected value of X_1 if $X_0 = 200$.

- 0.1
- 0.01
- 0.99
- 2

Question 9

If $dX_t = 2dt + 3dW_t$ and $dY_t = 3dt - 7dW_t$, then

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$d[X,Y]_t$ is:

- $-4dt$
- $-21dt$
- $3dt$
- $4dt$

Question 10

In a trinomial tree with $\sigma = 0.3$, $T = 1$, and $S_0 = 100$, compute the payoff of a European call option with $K = 90$ and $N = 2$ at $t = 2$ in path {m,m}.

- 14.6
- 2.14
- 44.99
- 10

Question 11

If X_t follows an Ornstein-Uhlenbeck (OU) process with parameters $a = 1$ and $\sigma = 0.3$, compute the variance of X_{10}.

- 0.045
- 0.043
- 0.015
- 0.025

Question 12

X_t follows a GBM with $\mu = 0.2$. If $X_0 = 100$, compute the expected value of X_5.

- 149.18

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QUESTIONS

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

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- 100
- 271.83
- 122.14

Question 13

If $dX_t = dt + tdW_t$ and $Y_t = X_t^2$, then dY_t is equal to:

- $2X_t dt + tX_t dW_t$
- $(2X_t + t^2)dt + 2dW_t$
- $t^2 dt + 2tX_t dW_t$
- $(2X_t + t^2)dt + 2tX_t dW_t$

Question 14

In a trinomial tree with $\sigma = 0.3$, $T = 1$, and $S_0 = 100$, compute the value of a European call option with $K = 90$ and $N = 2$ at $t = 1$ in path {u}.

- 17.5
- 44.96
- 44.99
- 92.21

Question 15

Assuming in a trinomial tree that the sum of probabilities is equal to one and that the martingale condition holds, we can say that...

- there are only two possible sets of risk-neutral probabilities.
- there is a unique set of risk-neutral probabilities.

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- the market is arbitrage-free.
- the market is not well defined.

Question 16

If X_t follows an Ornstein-Uhlenbeck process with parameters $a = 10$ and $\sigma = 0.5$, compute the variance of X_1.

- 0.07
- 0.026
- 0.42
- 0.012

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